

Android TV Player Validation — Findings

Revision: 1 **Last modified:** 2026-06-29T16:30:00Z **Device:** MIBOX4 @192.168.0.214:5555 (Android TV, SDK 28) **Authority:** Operator mandate (physical proof all players work, no bluff) + §11.4.107 (AV liveness battery) + §11.4.117 (pixel-oracle) + §11.4.143 (real-user journey) + §11.4.138 (operator-escape → bluff-audit + permanent guard) + §11.4.102 (systematic-debug).

Operator action items (read first)

#	Item	Status
1	Video player crashed 100% on launch — release blocker	✓ FIXED + PROVEN on-device

RESOLUTION — video playback FIXED and re-validated on-device (§11.4.107 PASS)

The crash below was fixed (ExoTvPlayerActivity → ComponentActivity, commit 995b5853) and **re-validated on the real MIBOX4 with the rebuilt APK**. Captured §11.4.107 proof: - **Crash gone:** “Play Now” now opens ExoTvPlayerActivity (foreground, app alive) — where it crashed 100% to the launcher before. - **Real movie loaded:** player overlay shows duration **2:04:12** (the real movie length, parsed from the mkv container) — 11_playing_position.png. - **Genuine playback (§11.4.107(2) frame-advance oracle):** position advanced **00:03 → 04:36 → 04:38** across captures, seek bar progressed — the movie is decoding + advancing. (09_player_frame1.png 00:03 → 11_playing_position.png 04:36.) - **PAUSE control works (§UI/UX):** CENTER toggled the player play↔pause — button icon flipped || (playing) → ► (paused), position froze at 04:38 (12_after_pause.png). Full transport overlay present: prev / -10s / play-pause / +10s / next / seek bar / settings. - **Honest §11.4.107 boundary:** the video *pixels* are not screencap-able — the player draws to a SurfaceView/hardware overlay that screencap returns

black (the surface is NOT FLAG_SECURE; this is the hardware-overlay capture limitation the constitution acknowledges: “true photon FPS at the sink has no clean software oracle”). The player’s own position-counter + duration + play/pause state ARE the capturable playback oracle, and they prove genuine advancing playback. Pixel-level visual confirmation is operator_attended per §11.4.52 (the operator sees the movie on the panel; the software proves it advances).

Original CRITICAL FINDING (now fixed) — video playback was completely broken

What was tested (real user journey, §11.4.143)

Launched app → pixel-oracle login (§11.4.117) → home screen rendered with **real TMDb covers** (Captain America + Iron Man posters — the cover-fix arc PROVEN on-device, evidence PROOF_covers_render_captain_america_ironman.png) → D-pad to “001 - Captain America - The First Avenger” → detail screen (★7.0, real backdrop, “Play Now”) → selected “Play Now”.

The defect (FACT — captured crash, not a guess)

Play Now launches ExoTvPlayerActivity (TVNavigation.kt:153). It **crashes instantly** on onCreate and the app dies back to the TV launcher home screen. Captured logcat (player_crash_log.txt):

```
java.lang.IllegalStateException: You need to use a Theme.AppCompat theme (or descendant) with this activity.
    at androidx.appcompat.app.AppCompatActivity.createSubDecor
    at
com.catalogizer.androidtv.ui.player.ExoTvPlayerActivity.onCreate(E)
Process com.catalogizer.androidtv (pid 7315) has died: fore TOP
```

Root cause (§11.4.102)

- ExoTvPlayerActivity : AppCompatActivity() (line 40) → setContentView(...) (line 49) requires a Theme.AppCompat descendant theme.
- Manifest assigns it @style/Theme.CatalogizerTV.Player → parent Theme.CatalogizerTV → parent **Theme.Leanback** (NOT an AppCompatActivity descendant). → setContentView throws.
- The other two players (MediaPlayerActivity, VLCPlayerActivity) extend ComponentActivity() — they tolerate the Leanback theme. Only ExoTvPlayerActivity (the one “Play Now” uses) is incompatible. **So video playback has NEVER worked on**

this build — every user who presses Play is kicked to the launcher.

Why prior testing missed it (§11.4.138 operator-escape class)

The prior HelixQA campaign drove navigation but never actually pressed Play + verified the player rendered a frame (it reported “100% coverage, 0 crashes” because the crash happens in a SEPARATE activity that returns to the launcher — not an in-app ANR/crash the foreground monitor saw). The §11.4.107 mandate (drive the real journey, verify genuine playback, don’t trust “launched”) is exactly what exposed this. A permanent §11.4.135 regression guard is required as part of the fix.

Fix (in progress, TDD §11.4.43 + §11.4.115 RED-on-broken-artifact)

Change ExoTvPlayerActivity to extend ComponentActivity (matching the working MediaPlayerActivity/VLCPlayerActivity pattern). ComponentActivity supports setContentView. Then re-validate the full §11.4.107 battery on-device.

Proven working (captured evidence)

- **Login** (§11.4.117 pixel-oracle): authenticated, catalog fetched (31 okhttp log lines).
- **Cover rendering** (the session’s cover-fix arc, proven on the real device): Captain America + Iron Man real TMDB posters on the home shelf —
PR00F_covers_render_captain_america_ironman.png.
- **Browse → detail journey**: D-pad navigation, detail screen with rating/backdrop/Play.
- **API stream endpoint**: /api/v1/entities/18/stream returns a real 4.6GB mkv stream_url; the API streams real bytes (HTTP 206, video/x-matroska; ~950KB/s warm).

Pending (after the player fix)

- Re-run §11.4.107 battery on the fixed player (freeze-oracle, decoder-health, UI controls).
- Audio (mp3) playback.
- Book/comic/image: build the missing readers (Phase 2, image-viewer + comic-pages API already built by parallel subagents this session), then validate.

Evidence files

File	Proves
PR00F_covers_render_captain_america_ironman.png	Real TMDB posters render on-device
06_after_select.png	Detail screen (real journey)
07_player_launched.png	Crash → back at TV launcher
player_crash_log.txt	The IllegalStateException theme crash
playback_attempt_recording.mp4	Window-scoped recording of the attempt